Homework 2

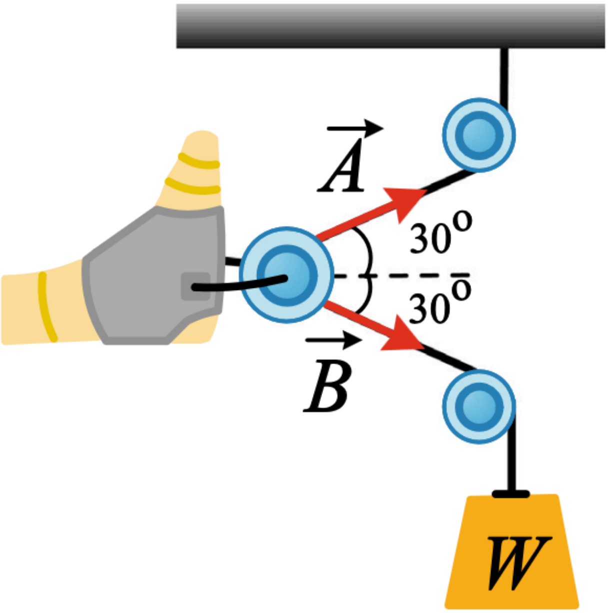
Due Tuesday, September 8, 2026 by 11:59 PM on Gradescope

Instructions

- Include your full name and Coyote ID on the top right of the first page
- Clearly circle or box your answer for any question involving calculations. If the problem is multipart, circle or box answers for each part.
- Work should be neatly written and easy to follow.
- You must tag each question on Gradescope. If you do not tag your work for a given problem, you will not receive credit for it.

Problems

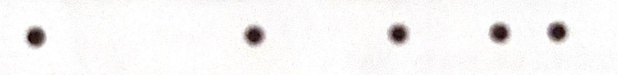
This homework contains some vector decomposition problems (material from week 1), motion graphs, as well as average velocity and and average acceleration in one-dimension.



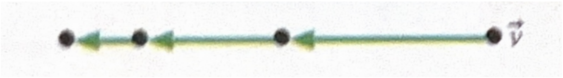
1. The figure above show s a traction apparatus which is used to prevent muscle contraction on a patient's leg. The pull of the two cords, which are presented by the vectors $\overrightarrow{A}$ and $\overrightarrow{B}$, have equal magnitude and combine to produce an outward horizontal traction force of $10\,\mathrm{N}$ on the patient's leg. Find the values of each pull (i.e., $\left|\overrightarrow{A}\right|$ and $\left|\overrightarrow{B}\right|$).

2. Convert the following to basic SI units or a combination of basic SI units (i.e., kilograms, meters, and seconds – kg, m, and s):
   (a) $8.0\,\mathrm{in}$
   (b) $66\,\mathrm{ft/s}$
   (c) $60\,\mathrm{mph}$
   (d) $14\,\mathrm{in}^2$

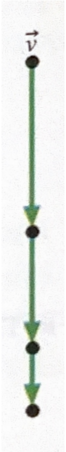
3. A runner on a straight track covers $1\ \mathrm{km}$ in $4\ \mathrm{min}$. What is their average velocity in
   (a) $\mathrm{km/min}$
   (b) $\mathrm{km/s}$
   (c) $\mathrm{km/h}$
4. Perform the following calculations with the correct number of significant figures:
   (a) 159.31×204.6
   (b) $5.1125 + 0.67 + 3.2$
   (c) $7.662 - 7.425$
   (d) $16.5/3.45$
5. The quantity called *mass density* is the mass per unit volume of a substance. What are the mass densities in basic SI units (i.e., kilograms, meters, and seconds — kg, m, and s) of the following objects:
   (a) $215\ \mathrm{cm}^3$ solid with a mass of $0.0179\ \mathrm{kg}$.
   (b) $95\ \mathrm{cm}^3$ liquid with a mass of $77\ \mathrm{g}$.



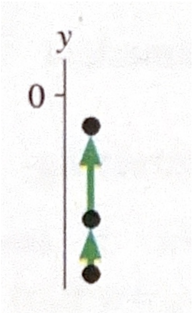
6. Is the particle in the figure above speeding up? Slowing down? Or can you tell? Explain.



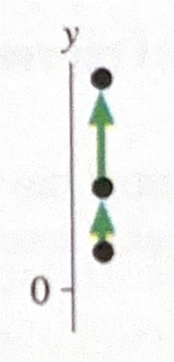
7. Does the object represented in the figure above have a positive or negative value of a_x? Explain.



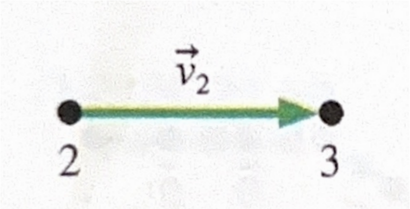
8. Does the object represented in the figure above have positive or negative value of a_y? Explain.



9. Determine the signs (positive, negative, or zero) of the position, velocity, and acceleration for the particle in the figure above.



10. Determine the signs (positive, negative, or zero) of the position, velocity, and acceleration for the particle in the figure above.
11. A baseball player stats running to the left to catch the ball as soon as the hit is made. Use the particle model to draw a motion diagram showing the position and average velocity vectors of the player during the first few seconds of the run.

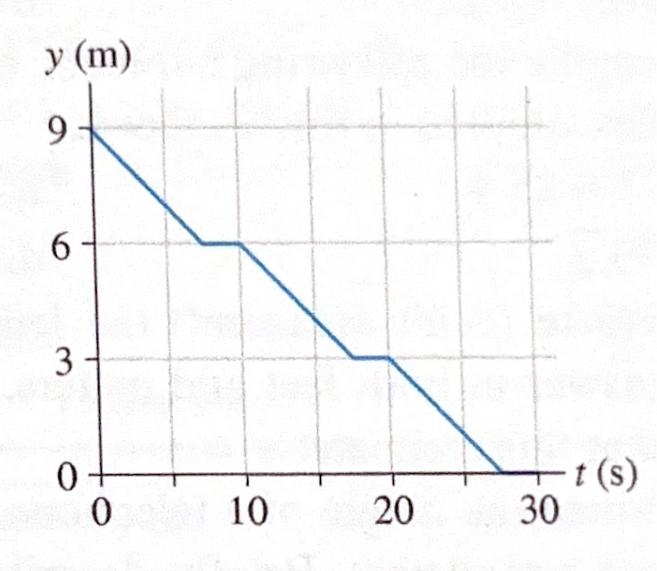


12. The figure above shows two dots of a motion diagram and vector $\vec{v}_2$.

    Copy this figure, then add dot 4 and the next velocity vector $\vec{v}_3$:

    (a) if the acceleration vector $\vec{a}$ at dot 3 points **right**

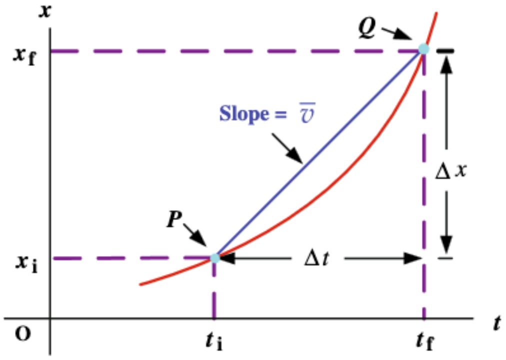
    (b) if the acceleration vector $\vec{a}$ at dot 3 points **left**



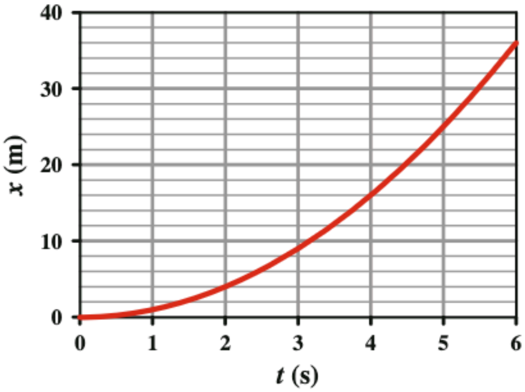
13. Write a short description of the motion of a real object for which the figure above would be a realistic position versus time graph.
14. A car travels in the positive x-direction for $20\,\mathrm{km}$ at $40\,\mathrm{km/h}$. It then continues in the same direction for another $20\,\mathrm{km}$ at $80\,\mathrm{km/h}$.

    (a) What is the average velocity of the car during this $40\,\mathrm{km}$ trip?

    (b) What is the average speed?



15. Suppose the motion of the particle in the above figure is described by the equation $x = a + bt^2$, where $a = 10\,\mathrm{m}$ and $b = 2\,\mathrm{m}/s^2$.
    (a) Find the displacement of the particle in the time interval between $t_i = 2\,\mathrm{s}$ and $t_f = 4\,\mathrm{s}$.
    (b) Find the average velocity and the average speed during this interval of time.



16. A graph of position (in meters) versus time (in seconds) for a person traveling in the positive x-direction is displayed in the figure above. Find the average velocity for the following cases:
    (a) $t_i = 2\,\mathrm{s}$ and $t_f = 4\,\mathrm{s}$
    (b) $t_i = 2\,\mathrm{s}$ and $t_f = 6\,\mathrm{s}$
    (c) $x_i = 12\,\mathrm{m}$ and $x_f = 30\,\mathrm{m}$

17. A body moves along a straight line with position even by $x = 8t - 2t^2$, where x is in meters and t is in seconds. Fine the average velocity and average speed of the body in the intervals:
    (a) From $t_i = 0$ to $t_f = 2\,\mathrm{s}$
    (b) From $t_i = 0$ to $t_f = 5\,\mathrm{s}$